

# Jagger Alexander

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## Education

- 2016 - 2020*     **B.A. in Mathematics**, *Vanderbilt University – Nashville, TN*  
Minor: Earth & Environmental Sciences | College of Arts & Sciences Honors Program
- 2020 - 2021*     **M.S. in Climate and Health**, *University of Miami – Miami, FL*  
Research Fellowship Award Recipient | GPA 4.0 / 4.0
- 2022 - 2025*  
*(expected May)*     **Ph.D. Candidate in Geosciences**, *University of Texas at Austin – Austin, TX*  
Jackson School of Geosciences Fellowship | GPA 4.0 / 4.0
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## Personal Statement

As a researcher and educator at the intersection of climate science and public health, I focus on understanding the health impacts of climate extremes. My research integrates global climate models, machine learning techniques, and population data to enhance climate predictions in ways which better inform public health interventions. My work has ranged from predicting the epidemiological impacts of weather on Zika outbreaks, to surveying physicians on how climate change is impacting their patients. I am also a certified Emergency Medical Technician (NREMT) and am committed to using my skills to serve communities disproportionately affected by climate change, translating scientific findings into actionable solutions for vulnerable populations.

## Professional Experience

- Fall 2024 - Spring 2025*     **Research Assistant**, University of Texas Marine Science Institute, Austin, TX
- Funded through a grant to characterize extreme weather events in Texas and their impacts on the Texas-Gulf Region.
  - Analyzed terabytes of novel climate data products, such as CONUS404, utilizing advanced supercomputing resources.
  - Collaborated with faculty and researchers focused on ecology and fisheries at UTMSI to provide expertise in analyzing time series data with climate covariates.
- Summer 2023*     **PREP-GC Intern**, Global Systems Laboratory, NOAA, Boulder, CO
- Developed front-end code for visualizing social vulnerability alongside natural hazard impact information in a web-based weather visualization system.
  - Processed, cleaned, and aggregated data from over two decades of natural hazards across CONUS, creating an application which enables point-radius searches. This is available at <http://sites.gsl.noaa.gov/DESI>.
- Summer 2020 – Fall 2022*     **Research Team Leader (Project MEER)**, Rowland Institute at Harvard, Cambridge, MA
- Led interdisciplinary teams comprising undergraduate and graduate students from diverse universities across the nation in the fields of terrestrial ecology, marine ecology, and energy modeling
  - Mentored undergraduates, providing guidance on proficient research techniques
  - Programmed a comprehensive model of incoming solar radiation for assessing the potential of terrestrial and floating marine mirror-arrays towards climate change mitigation

## Teaching Experience

- Spring 2024*     **Climate: Past, Present, and Future**, University of Texas at Austin, Austin, TX
- Independently designed and implemented lab experiments and assignments for a climate change course, integrating computer simulations, classroom experiments, and lectures to engage non-major undergraduates
  - Enhanced student engagement by incorporating current events and encouraging critical thinking, particularly through discussions on controversial topics like artificial cloud seeding, which successfully drew quieter students into active participation
- Fall 2023*     **Calculus Tutorial for Geoscience**, University of Texas at Austin, Austin, TX
- Designed and delivered curricula for three sections of a calculus tutorial, connecting mathematical concepts to geoscience topics like climate, hydrology, and geology
  - Independently managed all course duties, including syllabus creation, grading, and student support, while ensuring consistent classroom expectations
  - Fostered a supportive learning environment by building trust with students, offering personalized guidance, and connecting them with research and career opportunities aligned with their interests

## Volunteering

- Fall 2024 - Present*     **Longhorn EMS**, Austin, TX
- Volunteered at city wide events as a responder one, utilizing full scope of EMT-B practice
  - Completed relevant monthly trainings and participated in general meetings
- Summer 2024 - Present*     **National Alliance on Mental Illness (NAMI)**, Austin, Texas
- Undergoing training to teach short courses on mental health and facilitate support groups for individuals struggling with mental illness and their caregivers
- Summer 2018*     **Big Blue Conservation**, Koh Tao, Thailand
- Leveraged advanced open water diving certification to conduct marine censuses and construct artificial underwater reefs, contributing to efforts in mitigating climate change-related damage in the Gulf of Thailand
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## Publications

1. **Jagger Alexander** and Zong-Liang Yang (2025). Decoding sub-seasonal predictors of extreme heat with interpretable machine learning. <https://doi.org/10.31223/X5142V> (In review at *Artificial Intelligence for Earth Systems*)
2. Yamin Qing, Shuo Wang, Zong-Liang Yang, Pierre Gentine, Boen Zhang, and **Jagger Alexander**. Accelerated soil drying linked to increasing evaporative demand in wet regions. *npj Clim Atmos Sci* 6, 205 (2023). <https://doi.org/10.1038/s41612-023-00531-y>
3. **Jagger Alexander**, et al. (2022). Using machine learning to understand microgeographic determinants of the Zika vector, *Aedes aegypti*. *PLOS ONE* 17(12): e0265472. <https://doi.org/10.1371/journal.pone.0265472>
4. Marca Alexander, **Jagger Alexander**, Mohit Arora, Chloe Slocum, James Middleton (2019). A bellweather for climate change and disability: educational needs of rehabilitation professionals regarding disasters and spinal cord injuries. *Spinal Cord Series and Cases* 5(85). <https://doi.org/10.1038/s41394-019-0239-z>

5. Larisa R. G. DeSantis, **Jagger Alexander**, Eva M. Biedron, et al. (2019). Effects of climate on dental mesowear of extant koalas and two broadly distributed kangaroos throughout their geographic range. PLoS ONE 13(8): e0201962. <https://doi.org/10.1371/journal.pone.0201962>
6. Marca Alexander, Conley Carr, **Jagger Alexander**, Yuying Chen, Amie McLain (2018). Assessing the ability of the Sacral Autonomic Standards to document bladder and bowel function based upon the Asia Impairment Scale. Spinal Cord Series and Cases 5(85). <https://doi.org/10.1038/s41394-019-0228-2>.

## Presentations

1. **Jagger Alexander**, Zong-Liang Yang (2025). Extending heat wave forecasts with interpretable machine learning. Presented at American Meteorological Society Annual Meeting in New Orleans, LA.
  2. **Jagger Alexander**, Rob Howlett, and Travis Wilson (2023). Visualizing natural hazard vulnerability and impact. Presented at Climate and Health Conference 2023 at Zucker School of Medicine in Hempstead, NY. Also presented at American Geophysical Union Fall Meeting in San Francisco, CA.
  3. **Jagger Alexander** (2022). Weather Conditions and COVID-19 Incidence in Florida – Machine Learning and Classical Approaches. Presented at University of Miami Climate and Health Symposium in Miami, FL.
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## Skills

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|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Certifications</b> | <ul style="list-style-type: none"><li>• Emergency Medicine Technician (National Registry and Texas Certified)</li><li>• CPR and BLS</li><li>• Advanced Open Water Diver with 80+ dives</li></ul>                                    | <b>Geospatial</b>            | <ul style="list-style-type: none"><li>• ArcGIS</li><li>• GIS Technology</li></ul>                                                                                              |
| <b>Software</b>       | <ul style="list-style-type: none"><li>• JavaScript</li><li>• HTML</li><li>• Selenium</li><li>• Python</li><li>• R</li><li>• Microsoft Office Suite</li></ul>                                                                        | <b>Laboratory Techniques</b> | <ul style="list-style-type: none"><li>• Biology</li><li>• Chemistry</li><li>• Organic Chemistry</li></ul>                                                                      |
| <b>Data analysis</b>  | <ul style="list-style-type: none"><li>• Machine learning (random forest, gradient boosting, clustering)</li><li>• Time series analysis</li><li>• Spatial data interpolation (kriging, etc.)</li><li>• Probability testing</li></ul> | <b>Field Research</b>        | <ul style="list-style-type: none"><li>• Vertebrate Species Identification</li><li>• Field Counting Methods</li><li>• Mist-Netting</li><li>• Compass &amp; Navigation</li></ul> |

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## Honors

Jackson School of Geosciences Graduate Fellowship - Cornelius Vanderbilt Scholarship Recipient  
UMiami Climate and Health Graduate Fellowship - Weeks Climate Fellowship Honorable Mention  
National Merit Scholarship Recipient

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